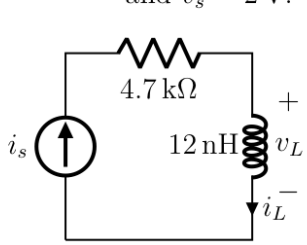
EE2015: Electric Circuits and Networks

Tutorial 1

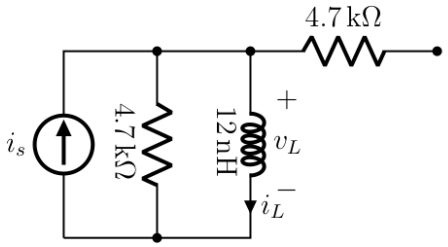
(August 8, 2025)

1. (a) For the circuits given in (i) to (iv), calculate v_L and i_L. Assume $i_s = 1\,\text{mA}$ and $v_s = 2\,\text{V}$.

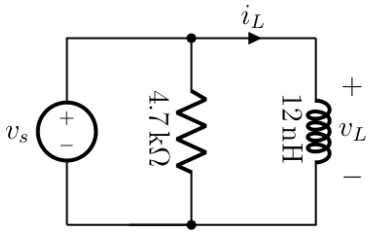
   (b) For the circuits given in (v) and (vi), find out the currents i_C. Assume $i_s = 1\,\text{mA}$ and $v_s = 2\,\text{V}$.



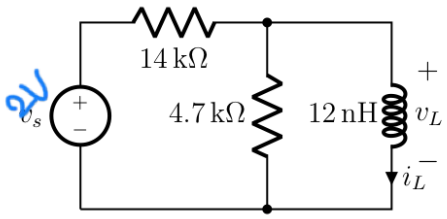
(i)



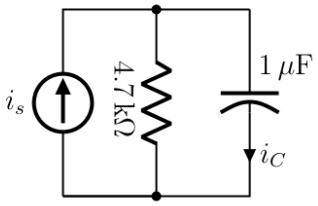
(ii)



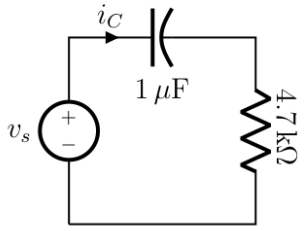
(iii)



(iv)



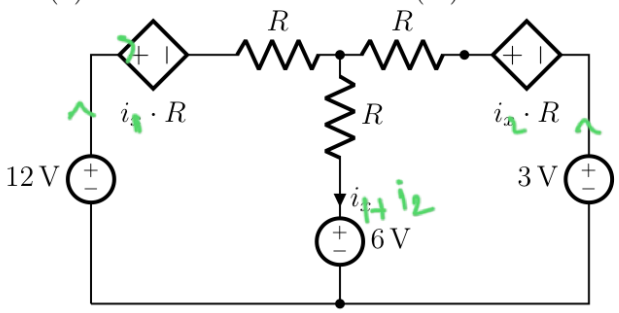
(v)



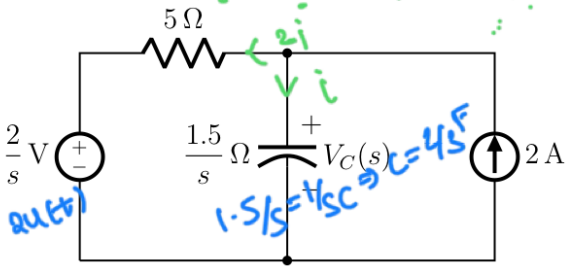
(vi)

2.

In the circuit shown on the right, determine the power *delivered* by the 6 V source when $R = 10\,\text{k}\Omega$.



3.



Consider the circuit shown on the left.

(a) Calculate $V_C(s)$.

(b) Determine $v_C(t)$ for $t \geq 0$.

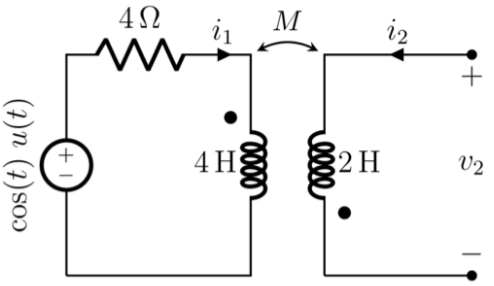
(c) Draw the time-domain representation of the circuit.

		V_L	i_L		i_C
Q1)	(i)	0	1mA	(v)	0mA
	(ii)	0	1mA	(vi)	0mA
	(iii)	2V	∞		
	(iv)	0	1/7mA		

5. Consider the circuit shown on the right, where the mutual inductance $M = 0.4\,\text{H}$.

   (a) Write the differential equation for $i_1(t)$. Assuming zero initial conditions, solve the differential equation using the Laplace transform approach.

   (b) Find $v_2(t)$.



$$\cos(t)u(t) = 4i_1(t) + \frac{4di_1(t)}{dt}$$

$$\frac{e^{jt}+e^{-jt}}{2}u(t) = 4I_1(s) + 4sI_1(s)$$

$$= \frac{1}{s+j} + \frac{1}{s-j} = 8I_1(s) + 8sI_1(s)$$

$$\Rightarrow \frac{1}{(s+j)(s+1)} + \frac{1}{(s-j)(s+1)} = 8I_1(s)$$

$$= \frac{1}{1-j}\left\{\frac{1}{s+j} - \frac{1}{s+1}\right\} + \frac{1}{1+j}\left\{\frac{1}{s-j} - \frac{1}{s+1}\right\} = 8I_1(s)$$

$$\Rightarrow \frac{1}{1-j}\left\{\frac{s}{s+j} - \frac{s}{s+1}\right\} + \frac{1}{1+j}\left\{\frac{s}{s-j} - \frac{s}{s+1}\right\} = 8sI_1(s)$$

$$\Rightarrow 8\frac{dI_1(t)}{dt} = \frac{1}{1-j}\left[\cancel{\delta(t)} - je^{-jt} - \cancel{\delta(t)} - e^{-t}\right] + \frac{1}{1+j}\left[\cancel{\delta(t)} + je^{jt} - \cancel{\delta(t)} - e^{-t}\right]$$

$$= e^{-t}\left[\frac{1}{j-1} - \frac{1}{j+1}\right] - \frac{je^{-jt}}{1-j} + \frac{je^{jt}}{1+j}$$

$$= -e^{-t} - \cos t + \sin t$$

$$\Rightarrow 0.4\frac{dI_1(t)}{dt} = V_2 = -\frac{1}{20}\left[e^{-t} + \cos t - \sin t\right]$$

$$\Rightarrow I_1(s) = \frac{1}{8(1-j)}\frac{1}{s+j} + \frac{1}{8(1+j)}\frac{1}{s-j} - \frac{2}{8}\frac{1}{s+1}$$

$$\Rightarrow I_1(t) = \frac{1}{8(1-j)}e^{-jt}u(t) + \frac{1}{8(1+j)}e^{jt}u(t) - \frac{1}{4}e^{-t}u(t)$$